9 To cool down the electrical generator of a nuclear power plant, cold water enters the heat exchanger of the generator at 3 °C and leaves at 11 °C. The rate of heat removed by the water is 4.0×10^{11} J per hour. The specific heat capacity of water is 4200 J kg^{-1} K^{-1}.

What is the rate of water flow?

A $\dfrac{4.0 \times 10^{11}}{4200 \times 8 \times 60 \times 60}$ kg s^{-1}

B $\dfrac{4.0 \times 10^{11} \times 60 \times 60}{4200 \times 8}$ kg s^{-1}

C $\dfrac{4.0 \times 10^{11}}{4200 \times 8 \times 60}$ kg s^{-1}

D $\dfrac{4.0 \times 10^{11} \times 60}{4200 \times 8}$ kg s^{-1}